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1. UART Specification

The UART (Universal Asynchronous Receiver/Transmitter) core provides serial communication capabilities, which allow communication with other external devices, like another computer using a serial cable and RS232 protocol. This core is designed to be maximally compatible with the industry standard National Semiconductors' 16550A device.

Features:

- APB interface in 32-bit data bus mode.
- FIFO only operation.
- Register level and functionality compatibility with NS16550A (but not 16450).
- Debug Interface in 32-bit data bus mode.
- No modem communication.

Register Description:

The original 16550 8-bit register bit fields are preserved but are mapped into 32 bit slots. The unused, 24 most significant, bits are read back as zeros.

Name	Address	Width	Access	Description
Receiver Buffer	0x0	8	R	Receiver FIFO output
Transmitter Holding Register (THR)	0x0	8	W	Transmit FIFO output
Interrupt Enable	0x4	8	RW	Enable/Mask interrupts generated by the UART
Interrupt Identification	0x8	8	R	Get interrupt information

FIFO Control	0x8	8	W	Control FIFO options
Line Control Register	0xc	8	RW	Control connection
Modem Control	0x10	8	W	Control modem
Line Status	0x14	8	R	Control information
Modem status	0x18	8	R	Modem Status
Divisor Latch1	0x1c	8	RW	16 bit Baud Rate divider – least significant byte
Divisor Latch2	0x20	8	RW	16 bit Baud Rate divider – most significant byte

UART Function Signals:

Signal	Width	Direction	Description
TXD	1	output	Serial transmit data output
RXD	1	input	Serial receive data input

Other Signals:

Signal	Width	Direction	Description
baud_o	1	output	Baud rate divider output
IRQ	1	output	Interrupt output

Register Description:

The original 16550 8-bit register bit fields are preserved but are mapped into 32 bit slots. The unused, 24 most significant, bits are read back as zeros.

Register	Offset	Width	Access	Description
RXD	0x0	8	RO	Receive FIFO output register
TXD	0x0	8	WO	Transmit FIFO input register
IER	0x4	8	R/W	Interrupt Enable Register
IIR	0x8	8	RO	Interrupt Identification Register
FCR	0x8	8	WO	FIFO Control Register
LCR	0xc	8	R/W	Line Control Register
MCR	0x10	8	R/W	Modem Control Register

LSR	0x14	8	RO	Line Status Register
MSR	0x18	8	RO	Modem Status Register
DIV1	0x1c	8	R/W	16 bit Baud Rate divider – least significant byte
DIV2	0x20	8	R/W	16 bit Baud Rate divider – most significant byte

Detailed Register Descriptions

TXD, RXD - Data buffer registers

Register	Bits	Access	Description
RXD	7:0	RO	Write to the bottom of the Transmit FIFO
TXD	7:0	WO	Read from the top of the Receive FIFO

IER – Interrupt Enable Register

This register enables the various interrupts available in the UART.

Bits	Access	Description
0	RW	Receive Data Interrupt enable
1	RW	Transmit Holding Register empty interrupt enable
2	RW	Receive Line Status interrupt enable
3	RW	Modem Status interrupt enable
7:4	RW	Reserved, will always read back as '0'

Reset Value: 0x0

Writing a logic '1' to any of these interrupt enable bits will enable the interrupt, and writing a logic '0' will disable the interrupt. After reset all interrupts are disabled.

IIR – Interrupt Identification Register

This read-only register is used to identify whether an interrupt is pending, and if so which interrupt.

Bit 0 of the register will read back as a logic '1' if no interrupt is pending, otherwise, the UART has an interrupt to be serviced. The valid codes returned in bits 3:0 of the register is described in the following table:

Code	Priority	Interrupt type	Interrupt source	Interrupt reset control
0x1	-	No interrupt	-	-
0x6	1 st	Receive Line status	Parity or Overrun or Framing error or Break Detected	Reading the line status register
0x4	2 nd	Receive data available	RX FIFO trigger level reached	RX FIFO drops below the trigger level
0xc	2 nd	Receive timeout	At least a character in the RX FIFO, but no character has been received by or read from the RX FIFO for at least 4 character periods	Reading from the receive data register
0x2	3 rd	Transmitter empty	The transmit FIFO and the transmit shift register is empty	Reading the IIR register or writing to the transmit data register

Reset Value: 0xC1

FCR – FIFO Control Register

The FIFO Control Register is a write-only register that allows the TX and RX FIFOs to be flushed. It also sets the level of the RX FIFO threshold.

Bits	Access	Description
0	W	Reserved
1	W	If set to logic '1', flushes the content of the RX FIFO, but does not affect the reception of any character that is in progress when the RX FIFO is flushed.
2	W	If set to logic '1', flushes the content of the TX FIFO, but does not affect any in progress transmission.
5:3	W	Reserved
7:6	W	Sets the RX FIFO interrupt threshold: '00' – 1 Character '01' – 4 Characters '10' – 8 Characters '11' – 14 Characters

Reset value: 0xc0

LCR – Line Control Register

The LCR controls the format of the serial characters transmitted and received by the UART.

Bits	Access	Description
1:0	RW	Defines the number of bits per character: ‘00’ – 5 bits ‘01’ – 6 bits ‘10’ – 7 bits ‘11’ – 8 bits
2	RW	Selects the number of stop bits transmitted: ‘0’ – 1 Stop bit ‘1’ – 2 Stop bits for 8,7,6 bit characters, 1.5 stop bits for 5 bit characters Note: The receive path only checks for 1 stop bit
3	RW	Parity Enable: ‘0’ – No parity ‘1’ – Parity enabled
4	RW	Even Parity Select: ‘0’ – Odd parity (Parity is set to ensure an odd number of 1’s in the overall character) ‘1’ – Even parity (Parity bit makes an even number of 1’s in the overall character)
5	RW	Stick Parity Select: ‘0’ – Stick parity disabled ‘1’ – The inverse of bit 4 is transmitted as the parity bit
6	RW	Break Control bit ‘0’ – Break disabled ‘1’ – The TX serial data line is forced to logic ‘0’ to indicate a break condition
7	RW	Reserved

Reset Value: 0x3

MCR – Modem Control Register

The MCR register provides the software interface to the modem control lines(we are not implementing), it also allows the UART loopback mode to be enabled.

Bits	Access	Description
3:0	RW	Reserved
4	RW	Loopback mode: '0' – Normal mode '1' – Loopback mode enabled
7:5	RW	Reserved

Reset Value: 0x0

LSR – Line Status Register

The LSR provides status information from the UART receive and transmit channels.

Bits	Access	Description
0	RO	Data Ready (DR) '0' – No characters in the RX FIFO '1' – At least one character in the RX FIFO
1	RO	Overflow Error (OE) The Overflow Error bit is set if a new character is received when the RX FIFO is full. The OE bit is cleared on reading the LSR
2	RO	Parity Error (PE) '1' – The character at the top of the RX FIFO was received with a parity error '0' – No parity error for the character at the top of the RX FIFO The PE bit is cleared on reading the LSR
3	RO	Framing Error (FE) '1' – The character at the top of RX FIFO was received with a framing error '0' – No framing error for the character at the top of the RX FIFO The FE is cleared on reading the LSR
4	RO	Break Interrupt (BI) A break interrupt is detected when the RXD serial input remains at logic '0' for at least one character period, one zero character is loaded into the RX FIFO and a breakinterrupt is flagged. The BI is cleared on reading the LSR
5	RO	Transmit FIFO empty '1' – The TX FIFO is empty – generates TX empty interrupt '0' – TX FIFO has at least one character in it The bit is cleared when a character is written to the TX FIFO
6	RO	Transmit Empty

		'1' – Both the transmit shift register and the TX FIFO are empty '0' – TX FIFO has at least one character, or the TX shift register is busy The bit is cleared when a character is written to the TX FIFO
7	RO	FIFO Error '1' – At least one parity error, framing error or break interrupt has been received and are stored in the RX FIFO. The bit is reset on reading the LCR. '0' – No errors

Reset Value: 0x60

DIV1 – DIVISOR LSB

The UART baud rate divider is implemented as a 16 bit counter which divides down the APB bus clock to arrive at a clock frequency which is 16x the serial data rate. The counter value is loaded via two 8 bit registers, DIV1 and DIV2.

Reading back from these registers will return the baud rate divider, not the current value of the divider counter.

The counter is started, or restarted, when the DIV1 register is written to.

Bits	Access	Description
7:0	RW	Least significant byte of the 16 bit baud rate divisor

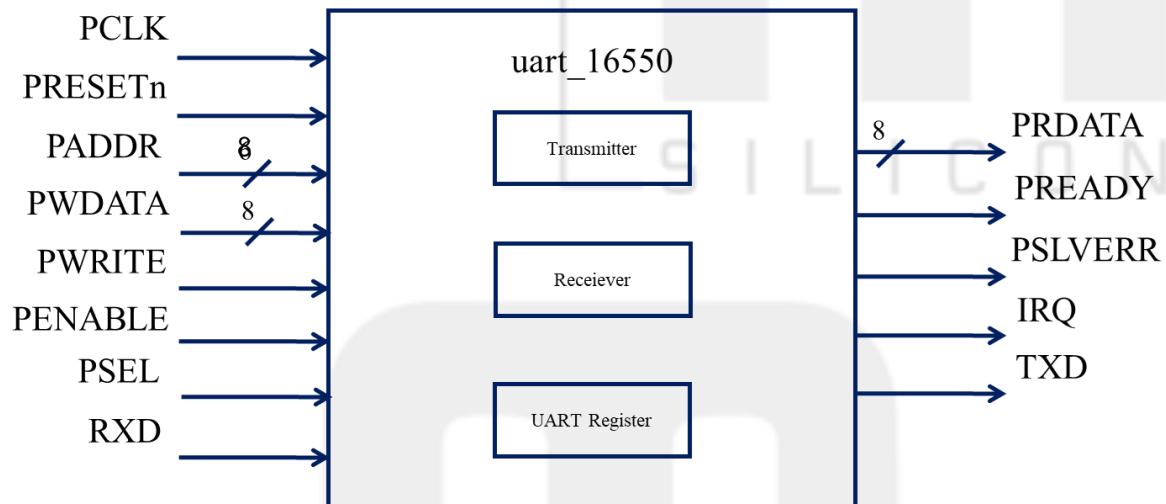
Reset value: 0x0

DIV2 – DIVISOR MSB

Bits	Access	Description
7:0	RW	Most significant byte of the 16 bit baud rate divisor

Reset value: 0x0

2. APB-based UART IP Core Block Diagram



Overview

The `uart_16550` module integrates a UART (Universal Asynchronous Receiver-Transmitter) interface compliant with the 16550 UART standard. It facilitates serial communication with an external system through APB bus signals and manages UART-specific functionalities such as transmission, reception, and interrupt handling.

Interface Signals

APB Bus Signals:

- PCLK: Clock signal for the APB bus.
- PRESETn: Active-low reset signal for the module.
- PADDR: Address bus for selecting registers.
- PWDATA: Data bus for writing data to registers.
- PRDATA: Data bus for reading data from registers.

- PWRITE: Write enable signal for indicating a write operation.
- PENABLE: Bus enable signal for indicating valid APB transactions.
- PSEL: Bus select signal indicating the module is selected.
- PREADY: Ready signal indicating the module is ready to accept or complete transactions.
- PSLVERR: Bus error signal indicating an error condition.

UART Specific Signals:

- TXD: Serial transmit data output.
- RXD: Serial receive data input.
- baud_o: Baud rate generator output, used for timing UART operations.
- IRQ: Interrupt request signal indicating an active interrupt condition.

Internal Signals

Transmitter Related:

- tx_fifo_we: Write enable for the transmit FIFO.
- tx_enable: Enable signal for transmitter operation.
- tx_fifo_count: Count of entries in the transmit FIFO.
- tx_fifo_empty: Transmit FIFO empty status.
- tx_fifo_full: Transmit FIFO full status.
- tx_busy: Status indicating the transmitter is actively transmitting data.

Receiver Related:

- rx_data_out: Data output from the receive FIFO.
- rx_idle: Status indicating the receiver is idle.
- rx_overrun: Receiver overrun error status.
- parity_error: Parity error status.

- `framing_error`: Framing error status.
- `break_error`: Break error status.
- `time_out`: Receiver timeout status.
- `rx_fifo_count`: Count of entries in the receive FIFO.
- `rx_fifo_empty`: Receive FIFO empty status.
- `push_rx_fifo`: Signal to push data into the receive FIFO.
- `rx_enable`: Enable signal for receiver operation.
- `rx_fifo_re`: Read enable for the receive FIFO.

Modem Interface Related:

- `loopback`: Loopback mode signal for testing purposes.

Modules and Interconnections

1. Control Module (`uart_register_file`):

- Manages register access and configuration based on APB bus transactions.
- Controls UART operations including transmit and receive data handling, interrupt management, and baud rate generation.

2. Transmitter Module (`uart_tx`):

- Handles transmit data path, including FIFO management and serial data transmission.
- Interfaces with the `uart_register_file` module for register access and configuration.

3. Receiver Module (`uart_rx`):

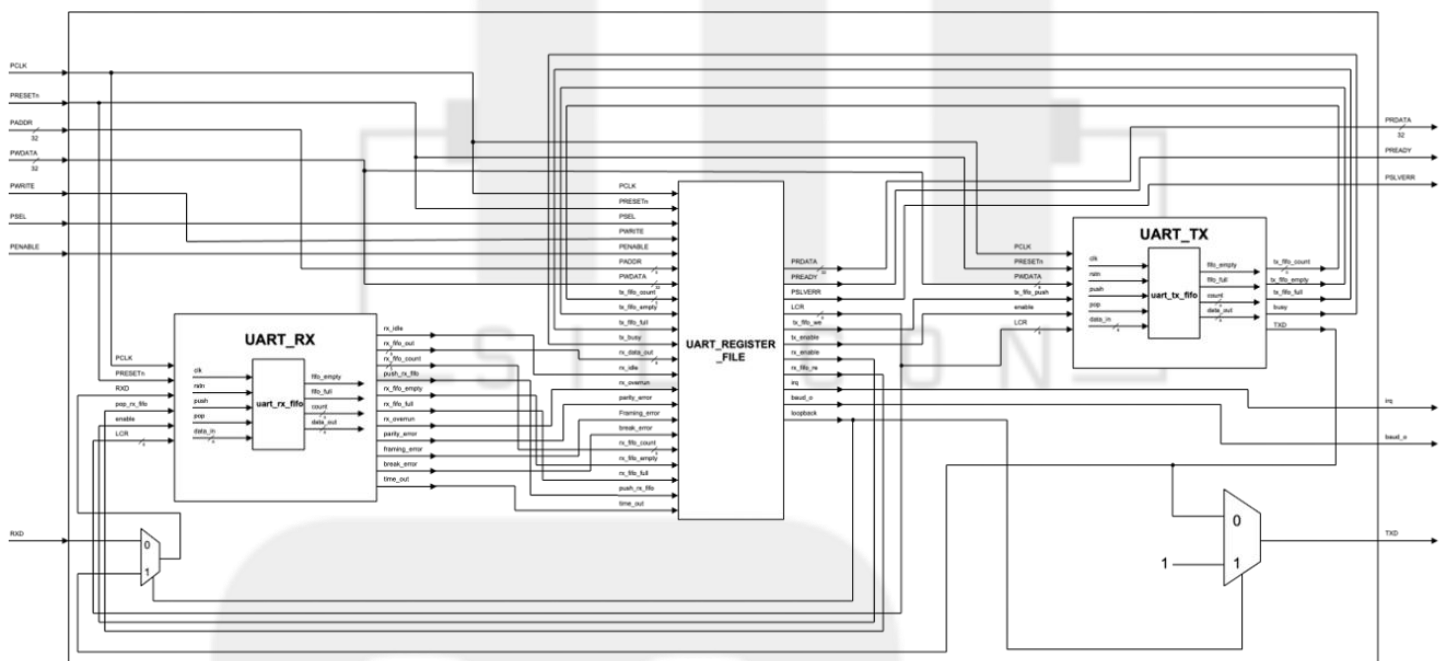
- Handles receive data path, including FIFO management and serial data reception.
- Interfaces with the `uart_register_file` module for register access and

configuration.

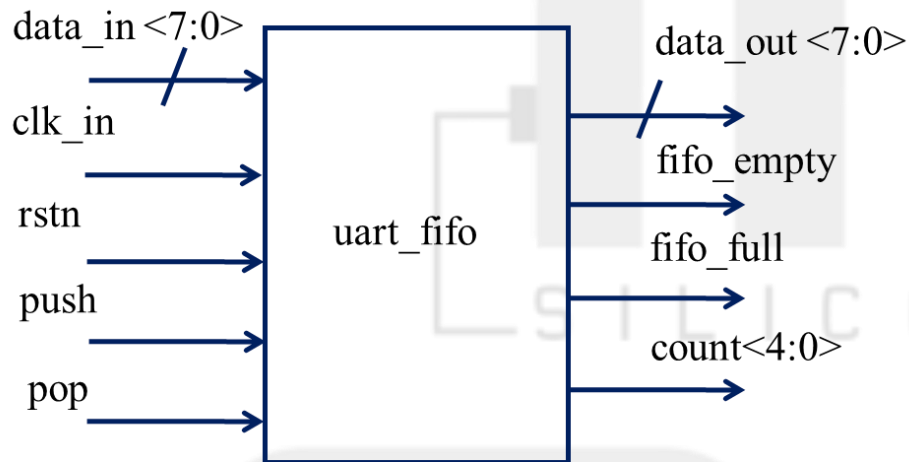
Loopback Handling

- RXDin and TXDout signals are interconnected to support loopback mode.
- When loopback is enabled, RXDin is connected to TXDout, allowing transmitted data to be looped back for testing purposes.
- Otherwise, RXDin connects directly to the external RXD input, and TXD outputs data received from TXDout

Top-level Microarchitecture

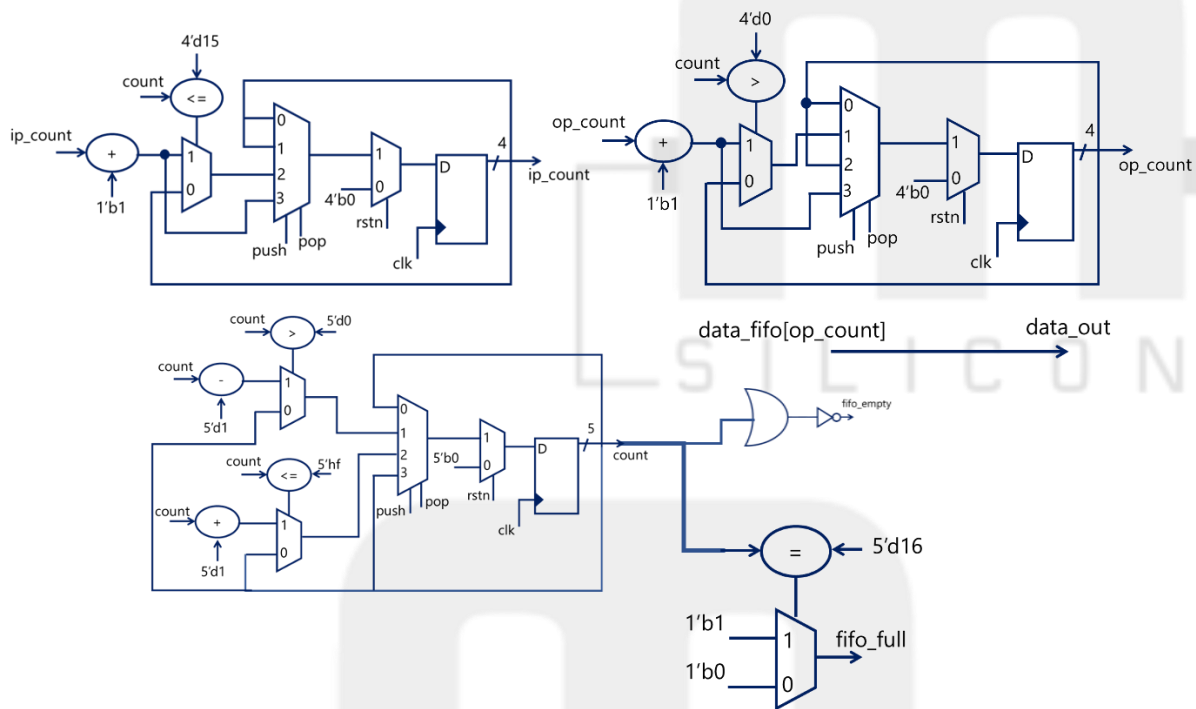


3. FIFO Block Diagram & Micro Architecture



Signal	Size	Direction	Function
clk_in	1	Input	Clock input
rstn	1	Input	Synchronous active low reset
data_in	8	Input	8-bit Data port to write data into memory
push	1	Input	Control signal to write into memory
data_out	8	Output	8- bit Data port to read from the memory
pop	1	Input	Control signal to read from the memory
fifo_empty	1	Output	Output signal to indicate if memory is empty
fifo_full	1	Output	Output signal to indicate if memory is full
count	5	Output	To indicate number of bytes written and read from the memory

Microarchitecture



Functionality

- UART has 2 FIFO memory one at the Transmitter and one at the receiver.
- The transmitter FIFO is an 8-bit wide and 16-location deep FIFO memory buffer. Data written across the APB interface is stored in the FIFO until read out by the transmit logic.
- The control signal for read operation from FIFO is generated internally from the transmitter FSM.
- The receive FIFO is a 8-bit wide and 16 location deep FIFO memory buffer. Received data and corresponding error bits are set by the receive logic until read out by the CPU across the APB interface.
- The control signal for write operation is generated internally from the reciever FSM.

Pseudocode:

1. Reset Behavior:

- When rstn is low (0), the FIFO is reset:count, ip_count ,op_count is set to 0.

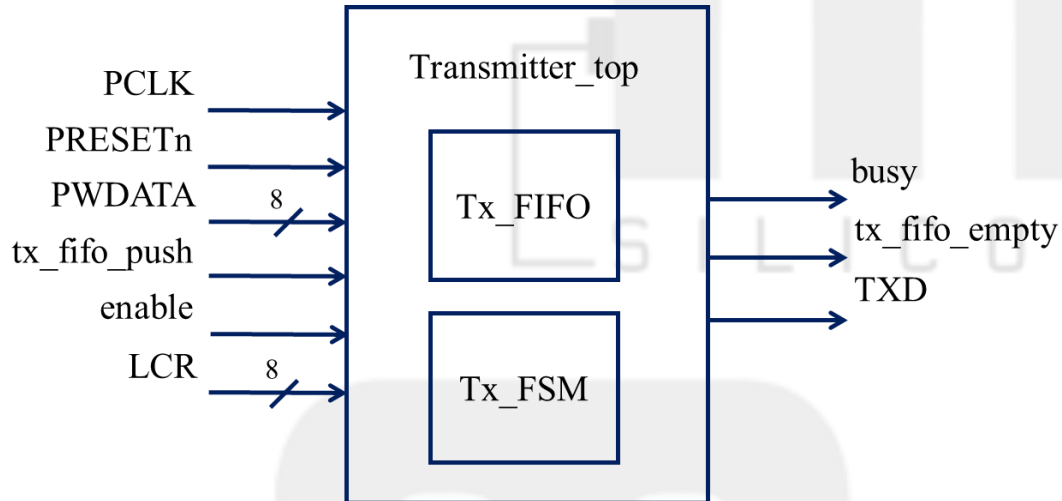
2. Push and Pop Operations:

- Push Only (push = 1, pop = 0): If the FIFO is not full (count ≤ 15), the data from data_in is written to the FIFO at the position indicated by ip_count. ip_count, count is incremented by 1.
- Pop Only (push = 0, pop = 1): If the FIFO is not empty (count > 0), the data at the position indicated by op_count is available at data_out. op_count, is incremented by 1 and count is decremented by 1.
- Simultaneous Push and Pop (push = 1, pop = 1): The data from data_in is written to the FIFO at the position indicated by ip_count. The data at the position indicated by op_count is available at data_out. Both ip_count and op_count are incremented by 1. count remains unchanged.

Combinational Logic

1. Data Output (data_out): The data at the position indicated by op_count in the FIFO is assigned to data_out.
2. FIFO Empty Signal (fifo_empty): fifo_empty is asserted (1) when count is 0, indicating the FIFO is empty.
3. FIFO Full Signal (fifo_full): fifo_full is asserted (1) when count is 16, indicating the FIFO is full.

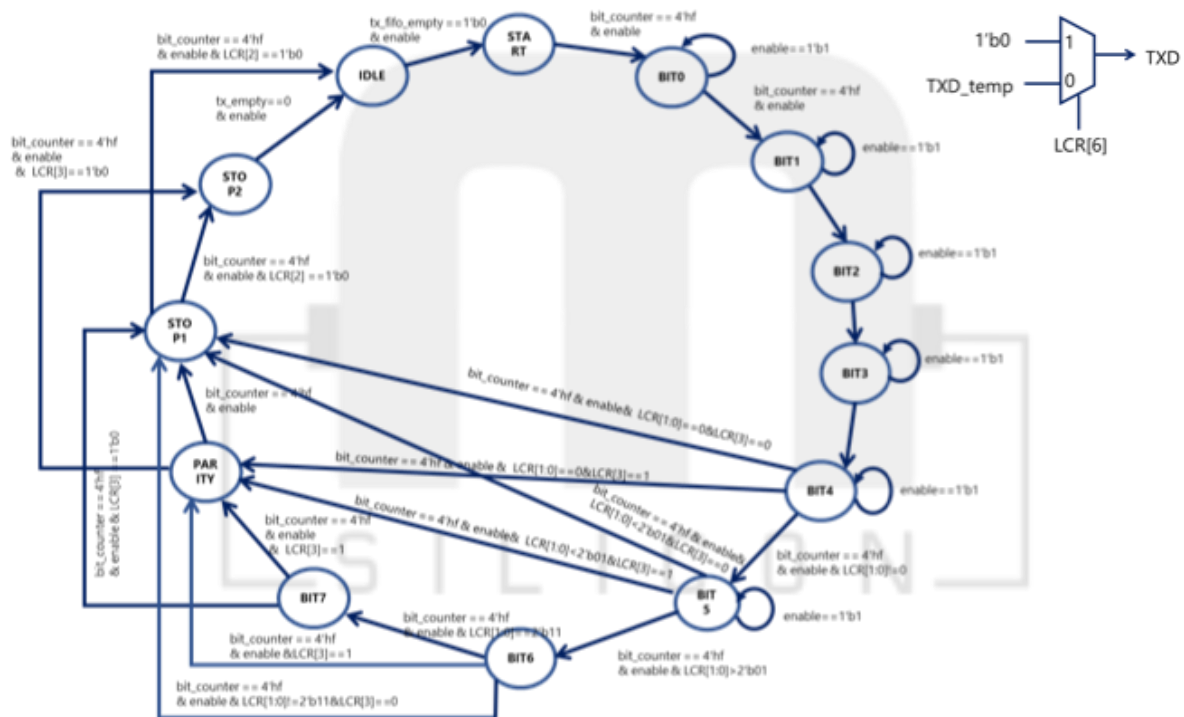
4. UART Transmitter Block Diagram & Micro Architecture



Signal	Size	Direction	Function
pclk	1	Input	Clock. The rising edge of PCLK times all transfers on the APB.
presetn	1	Input	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.
pwdata	8	Input	Write data. This bus is driven by the peripheral bus bridge unit during write cycles when PWRITE is HIGH. This bus can be up to 8 bits wide.
tx_fifo_push	1	Input	Control signal to write into transmitter memory
enable	1	Input	Baud pulse for serial data transmission
LCR	8	Input	8-bit input to load the contents of Line Control Register
tx_fifo_count	5	Output	5-bit output to indicate number of byte present in the memory

busy	1	Output	Will be high till data bits are transmitted
tx_fifo_empty	1	Output	Output signal to indicate if transmitter memory is empty
tx_fifo_full	1	Output	Output signal to indicate if transmitter memory is full
TXD	1	Output	1-bit output used to transmit data serially out

Transmitter Controller:



Functionality:

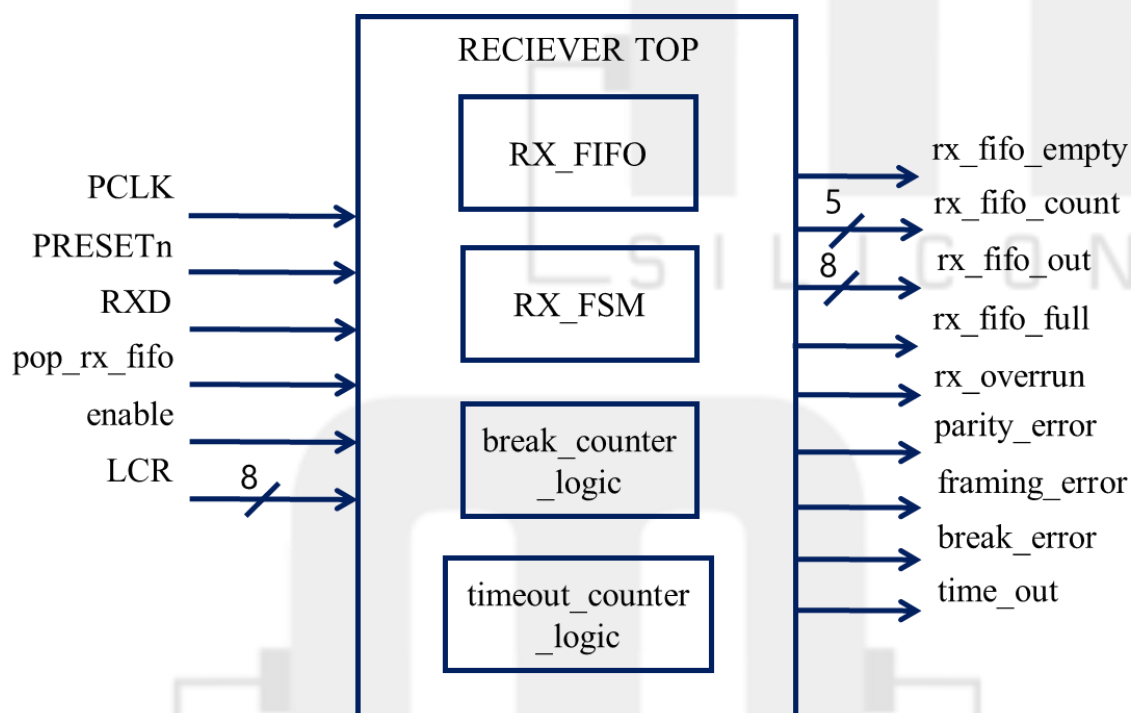
- **UART transmitter** is used to transmit the serial data to receiver module of other UART device.
- Transmitter will receive parallel data from APB interface and store it in Transmitter FIFO.
- After getting the data from the data bus, it will add a start bit, a parity bit, and a stop bit to create the character frame.
- The character frame is then transmitted at the TX pin

- The receiving UART will read the data packet at its RX pin. Data is sent until there is no data left in the transmitting UART.
- Internal Registers and Wires
 1. tx_state: 4-bit register representing the current state of the transmit FSM.
 2. bit_counter: 4-bit counter for tracking the number of clock cycles or bits
 3. transmitted in the current state.
 4. tx_buffer: 8-bit register holding the current byte to be transmitted.
 5. tx_fifo_out: 8-bit wire for reading data from the transmit FIFO.
 6. pop_tx_fifo: Signal to pop data from the transmit FIFO. Active high.
 7. TXD_tmp: Temporary register holding the current value to be transmitted on
 8. the TXD line.
- State Definitions
 1. IDLE (4'b0000): Idle state, waiting for data to transmit.
 2. START (4'b0001): Start bit transmission state.
 3. BIT0 (4'b0010) to BIT7 (4'b1001): States for transmitting data bits 0 to 7.
 4. PARITY (4'b1010): Parity bit transmission state.
 5. STOP1 (4'b1011): First stop bit transmission state.
 6. STOP2 (4'b1100): Second stop bit transmission state (if configured).

Pseudocode:

- Initialize UART Transmitter: - Set 'busy' flag to False (indicates transmitter is not busy) - Initialize Set 'transmit_shift_register' to 0 (used for serial transmission).
- Write data in FIFO memory.
- Start the transmission by asserting 'busy' flag and read data from FIFO to 'transmit_shift_register' make character frame with start,data bits,parity and stop bit.
- Each bit of the character frame to be transmitted using the Baud interval set by the clock signal

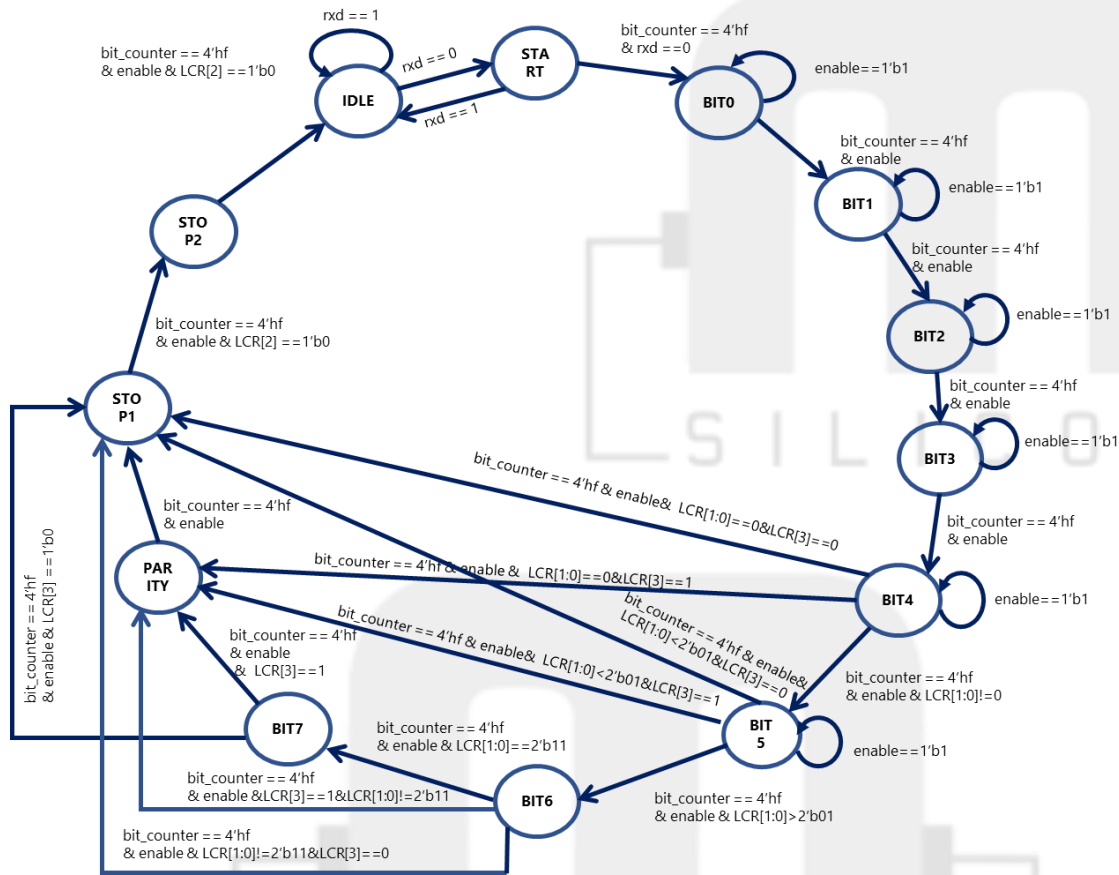
5. UART Reciever Block Diagram & Micro Architecture



Signal	Size	Direction	Function
PCLK	1	Input	Clock. The rising edge of PCLK times all transfers on the APB.
PRESETn	1	Input	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.
RXD	1	Input	Pin to receive serial data
pop_rx_fifo	1	Input	Control signal to read from reciever memory
enable	1	Input	Baud pulse for serial data transmission
LCR	8	Input	8-bit input to load the contents of Line Control Register
rx_idle	1	Output	

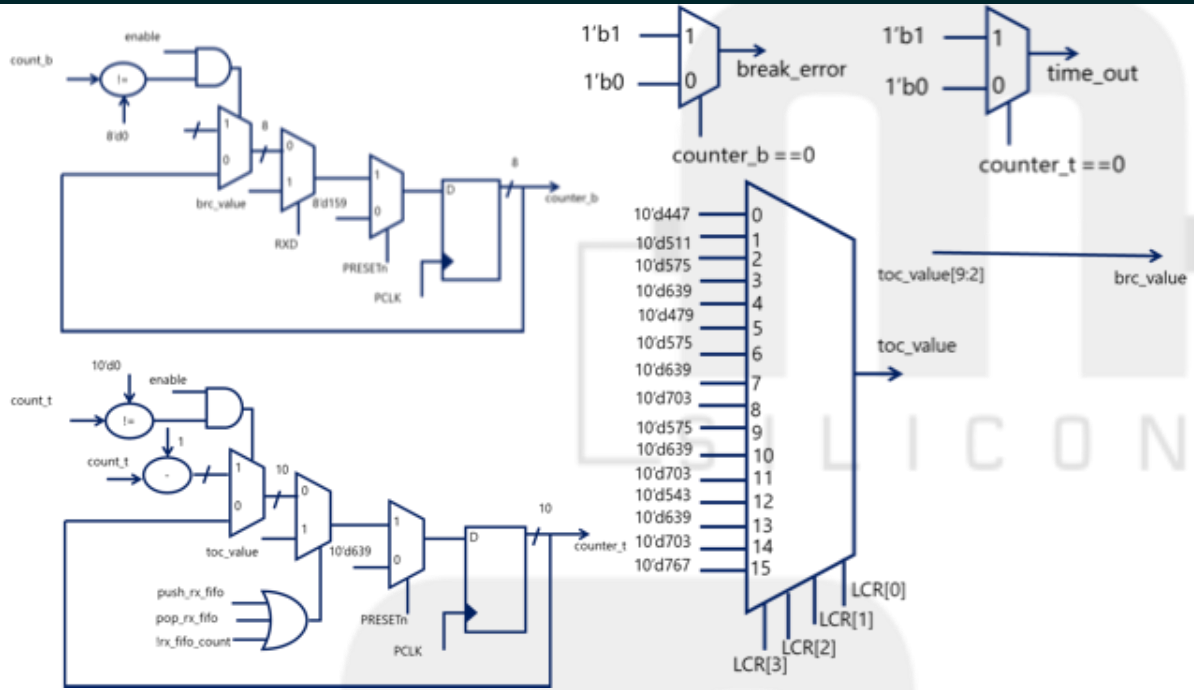
rx_fifo_out	8	Output	Number of bytes present in memory
rx_fifo_count	5	Output	5-bit output to indicate number of byte present in the memory
push_rx_fifo	1	Output	Control signal to write into receiver FIFO
rx_fifo_empty	1	Output	Output signal to indicate if receiver memory is empty
rx_fifo_full	1	Output	Output signal to indicate if receiver memory is full
rx_overnrun	1	Output	Overrun flag
parity_error	1	Output	Parity error
framing_error	1	Output	Framing error
break_error	1	Output	Break error
time_out	1	Output	Time error

Receiver Controller:

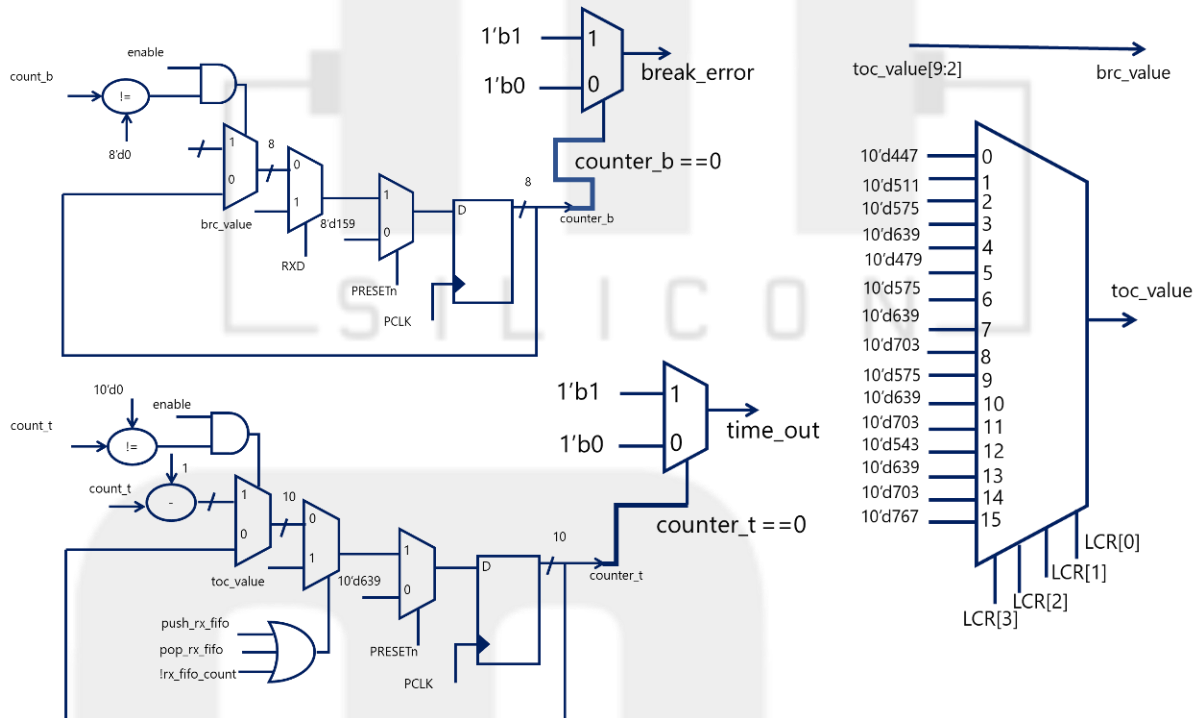


Microarchitecture:

Part1



Part2

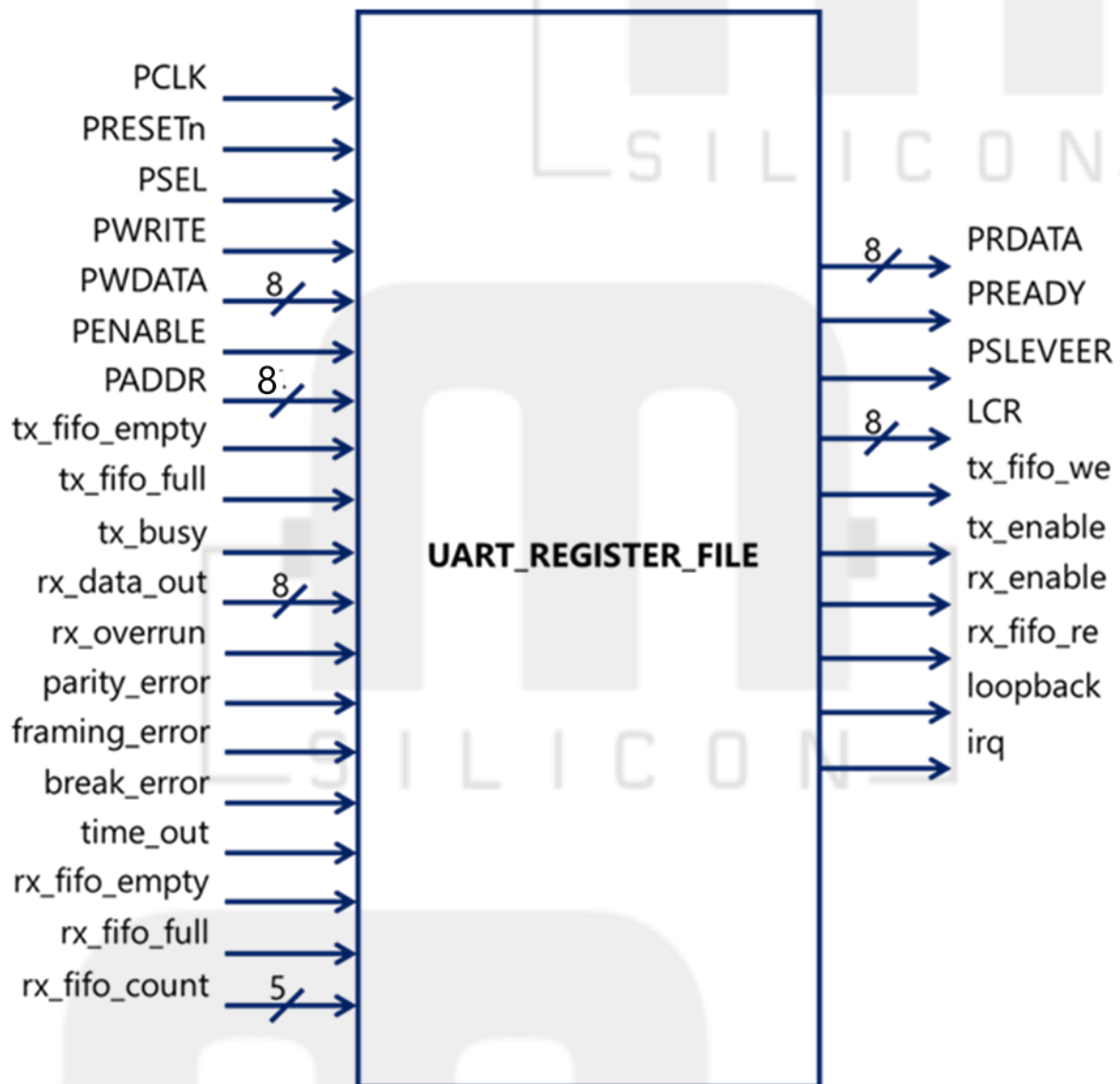


Functionality

- **UART reciever** is used recieve the serial data from the transmitter module of other UART device.
- Reciever will receive serial data.
- Reciever module converts the serial data to parallel data and store it in the receiver FIFO
- UART reciever_FIFO block is used to store the date that is available for serial to parallel conversion.
- The receive FIFO is a 8-bit wide and 16 location deep FIFO memory buffer. Received data and corresponding error bits are set by the receive logic until read out by the CPU across the APB interface.
- The control signal for write operation is generated internally from the reciever FSM.
- Internal Signals and Registers
 1. rx_state: Current state of the FSM (4 bits).
 2. bit_counter: Counter for tracking bits received (4 bits).
 3. rx_buffer: Buffer to store received bits (8 bits).
 4. brc_value: Break counter value (8 bits).
 5. framing_error_temp: To check the break error then update the framing
 6. error.
 7. toc_value: Timeout counter value (10 bits).
 8. counter_b: Break counter (8 bits).
 9. counter_t: Timeout counter (10 bits).
- FSM State Definitions
 1. IDLE: Waiting for the start bit.
 2. START: Validating the start bit.
 3. BIT0 to BIT7: Receiving data bits.
 4. PARITY: Checking the parity bit.
 5. STOP1, STOP2: Checking stop bits.

6. UART Register Block Diagram & Micro Architecture

Block diagram:



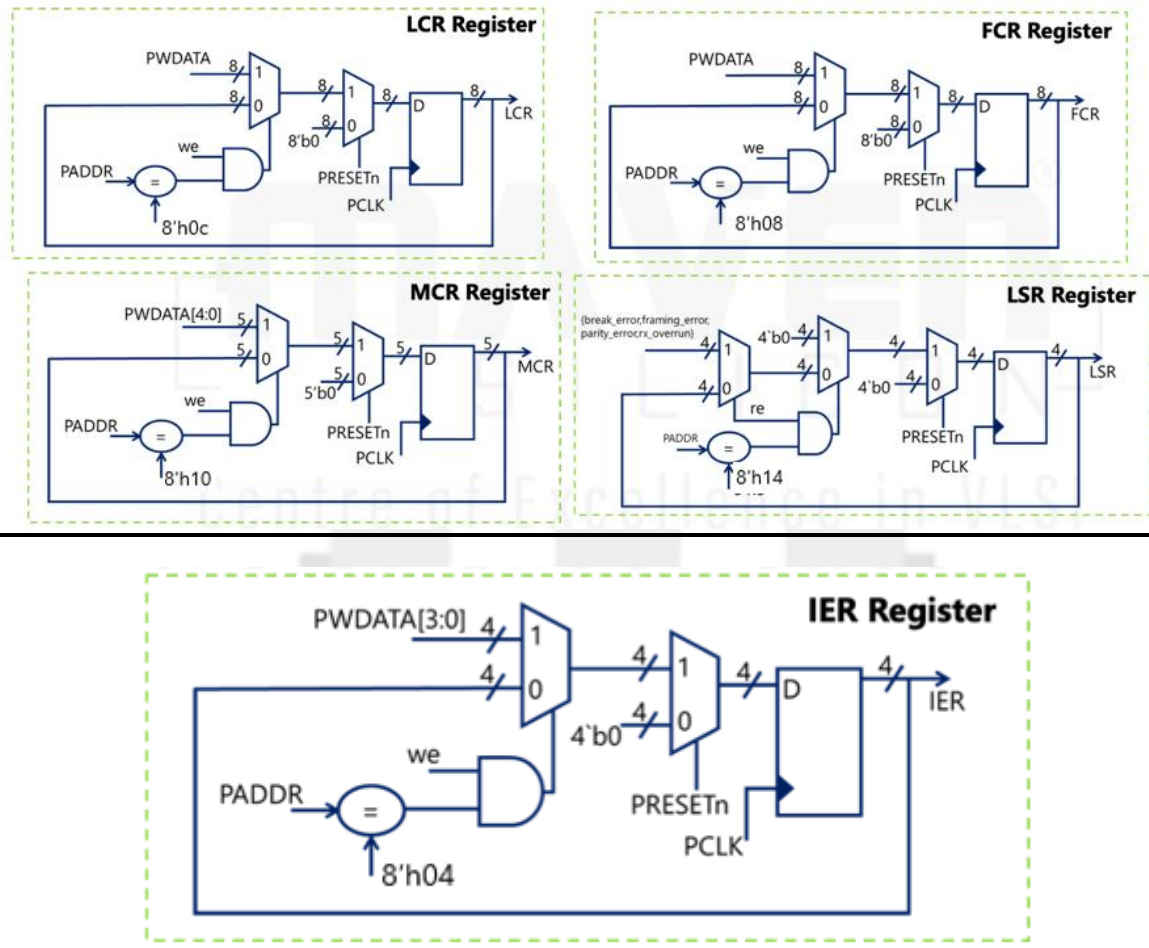
Signal Description:

Signal	Size	Direction	Function
PCLK	1	Input	Clock. The rising edge of PCLK times all transfers on the APB.
PRESETn	1	Input	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal.
PSEL	1	Input	Signal used to enable the UART module for access when active
PWRITE	1	Input	Indicates a write operation when active, specifying that data (PWDATA) should be written to the selected register (PADDR).
PENABLE	1	Input	Signal indicating an active access cycle to the UART registers.
PADDR	32	Input	Address bus specifying which register is being accessed within the UART module.
PWDATA	32	Input	Data bus carrying the information to be written into the selected register
tx_fifo_count	5	Input	Number of entries currently in the TX (transmit) FIFO.
tx_fifo_empty	5	Input	Indicates whether the TX FIFO is currently empty.
tx_fifo_full	1	Input	Indicates whether the TX FIFO is currently full
tx_busy	1	Input	tx_busy Status indicating whether the transmitter is actively transmitting data
rx_data_out	8	Input	Data output from the RX (receive) FIFO.
rx_idle	1	Input	Indicates whether the receiver is currently idle.
rx_overrun	1	Input	Indicates if a receiver overrun error has occurred
parity_error	1	Input	Indicates if a parity error has been detected in received data.
framing_error	1	Input	Indicates if a framing error has been detected in received data.
break_error	1	Input	Indicates if a break condition has been detected in received data
time_out	1	Input	Indicates if a timeout condition has occurred during operation.
rx_fifo_count	1	Input	Number of entries currently in the RX FIFO.

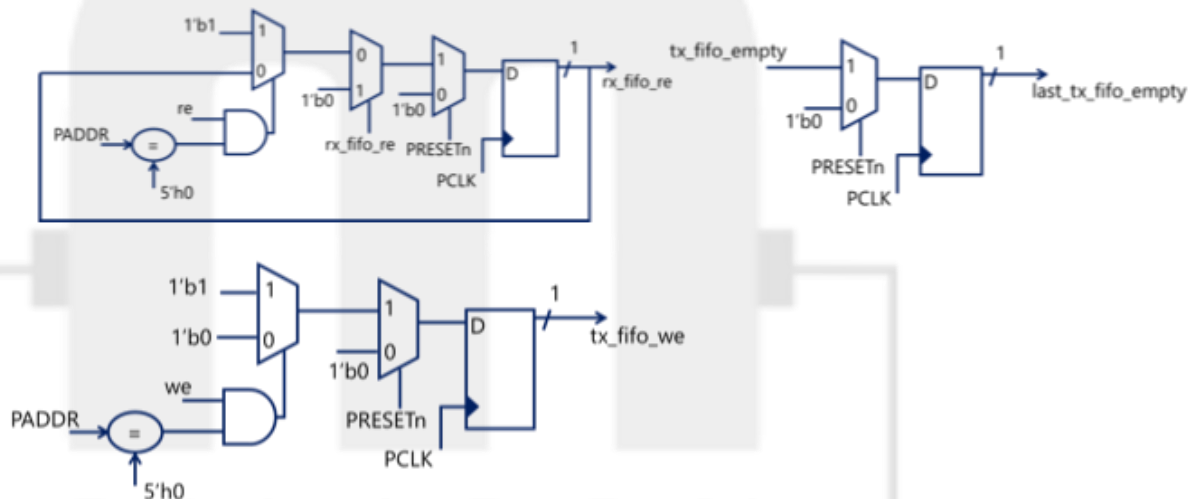
rx_fifo_empty	1	Input	Indicates whether the RX FIFO is currently empty.
rx_fifo_full	1	Input	Indicates whether the RX FIFO is currently full.
push_rx_fifo	1	Input	Signal to push data into the RX FIFO.
parity_error			Indicates if a parity error has been detected in received data.
framing_error	1	Input	Indicates if a framing error has been detected in received data.
break_error	1	Input	Indicates if a break condition has been detected in received data
time_out	1	Input	Indicates if a timeout condition has occurred during operation.
rx_fifo_count			Number of entries currently in the RX FIFO.
rx_fifo_empty	1	Input	Indicates whether the RX FIFO is currently empty.
rx_fifo_full	1	Input	Indicates whether the RX FIFO is currently full.
push_rx_fifo	1	Input	Signal to push data into the RX FIFO.
PRDATA		Output	Data output bus carrying the content read from the selected register (PADDR).
PREADY	1	Output	Ready signal indicating the completion of an access operation
PSLVERR	1	Output	Error signal indicating a bus error during an access cycle.
LCR	1	Output	Output register controlling line configuration settings for UART communication
tx_fifo_we	1	Output	Write enable signal for the TX FIFO, controlling when data can be written into the FIFO
tx_enable	1	Output	Enable signal controlling the transmission of data from the TX FIFO
rx_enable	1	Output	Enable signal controlling the reception of data into the RX FIFO
rx_fifo_re	1	Output	Signal to read (pop) data from the RX FIFO.
PRDATA	32	Output	Data output bus carrying the content read from the selected register (PADDR).
PREADY	1	Output	Ready signal indicating the completion of an access operation
loopback	1	Output	Output signal controlling the loopback mode for testing and diagnostic purposes
irq	1	Output	Output signal indicating various UART events that require attention or processing
baud_o	1	Output	Signal indicating the baud rate output for timing purposes during data transmission

Microarchitecture:

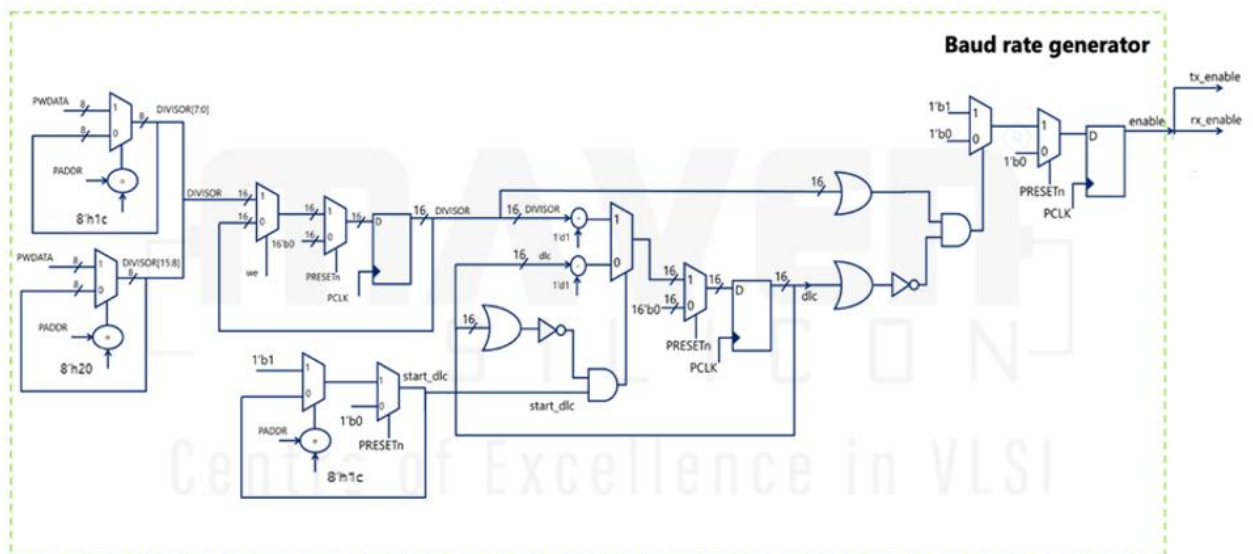
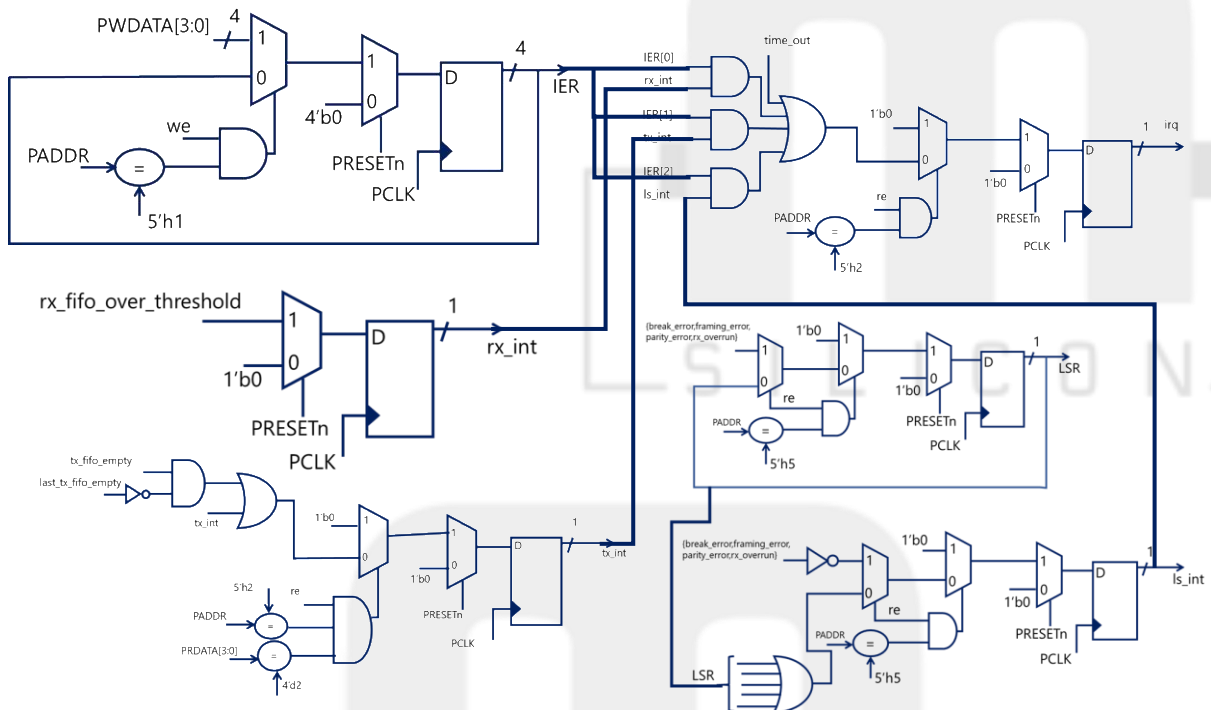
Part1



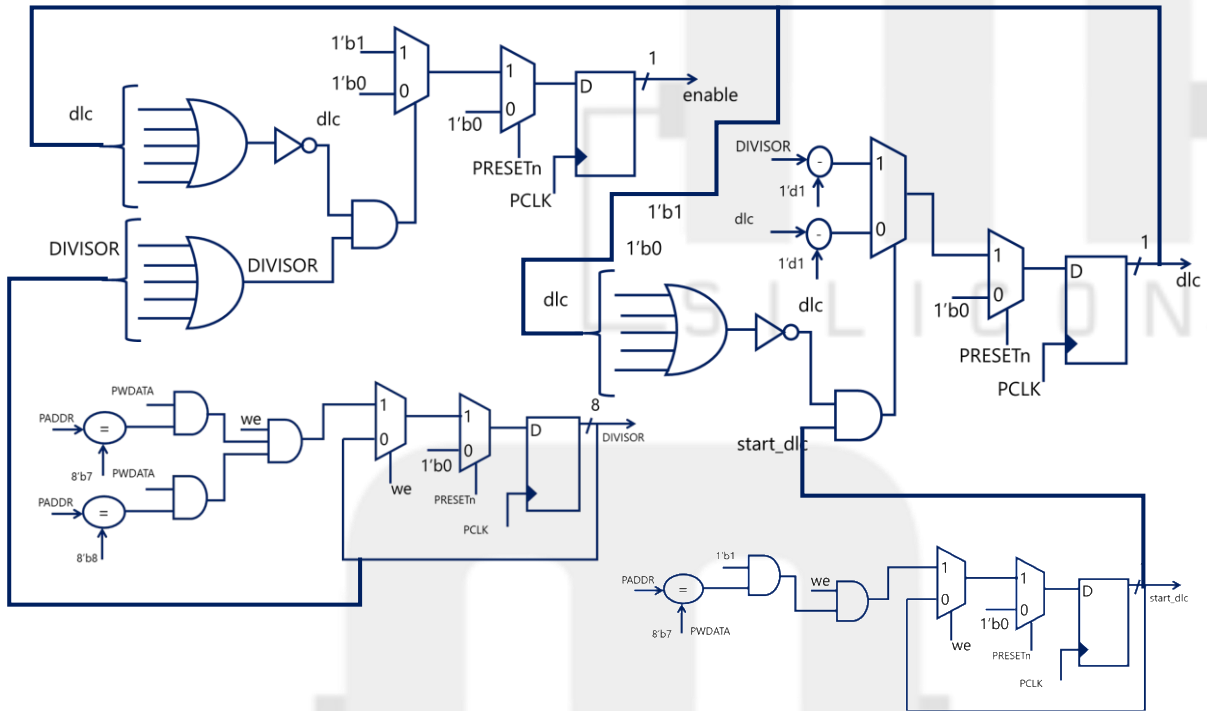
Part2



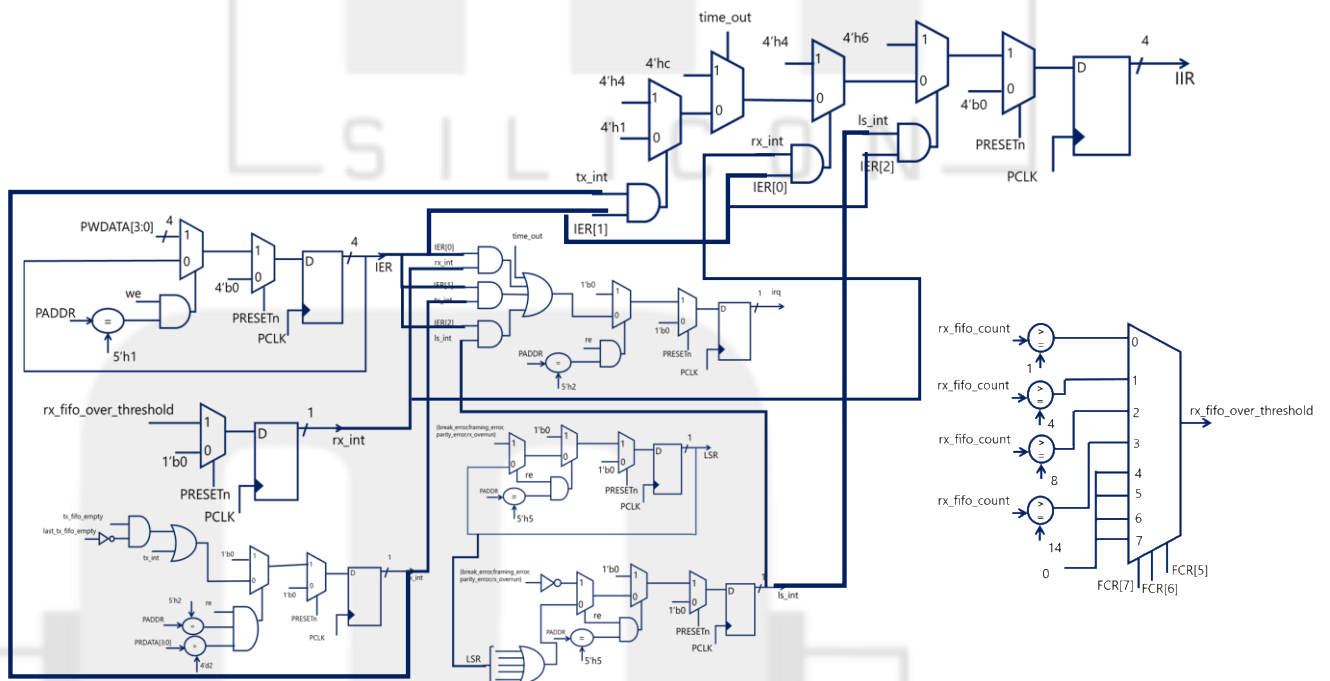
Part3



Part4



Part5



Functionality:

- This module is a UART (Universal Asynchronous Receiver/Transmitter) register file interfaced with an APB (Advanced Peripheral Bus).
- It manages control and status registers, handles data transmission and reception, and generates interrupts for various UART events.
- APB bus interface FSM: The FSM controls the APB bus interface for accessing UART registers. It manages read and write operations by transitioning through states based on the APB bus signals.

States

IDLE (2'b00)

Description: Default state where the FSM waits for a transaction.

Actions: we (write enable) is set to 0, re (read enable) is set to 0, PREADY is set to 0.

Transitions: If PSEL is asserted, transition to SETUP.

SETUP (2'b01)

Description: Prepares for a read or write operation.

Actions: re is set to 0.

Transitions: If PSEL and PENABLE are asserted, transition to ACCESS. If PWRITE is asserted, set we to 1. Else, transition back to IDLE and set we to 0.

ACCESS (2'b10)

Description: Executes the read or write operation.

Actions: PREADY is set to 1, we is set to 0. If PWRITE is not asserted, set re to 1.

Transitions: Transition back to IDLE.

DEFAULT

Description: Default state to handle unexpected conditions.

Actions: Transition back to IDLE.

